|INCH-POUND|

MIL-M-28787/512
AMENDMENT 1
29 September 1993

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MEMORY MAPPED I/O MODULE, KEY CODE BWG

This amendment forms a part of Military Specification MIL-M-28787/512, dated 15 September 1989, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 4

TABLE I, Test condition A: Add End-of-life limits below the current isolation test with a minimum limit of 1 MOhm (all temperatures) and a maximum limit of INF MOhm (all temperatures).

PAGE 6

TABLE I, Test condition column, Test condition J: Delete "J1-J42" and substitute "J1-J41".

PAGE 7

TABLE I, Test condition R12, Max column: Delete "50" (three places) and substitute "75" (three places).

TABLE I, Test condition R39, Max column: Delete "50" (three places) and substitute "75" (three places).

PAGE 20

6.3.1, Signal "AD ID B11", Cin column: Delete "50" and substitute "75".

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PAGE 23

6.3.1, Signal "EXT ID B3", Cin column: Delete "50" and substitute "75".

PAGES 28 THROUGH 34

6.3.3: Delete and substitute new 6.3.3 as printed on pages 4 through 11 of this amendment.

PAGES 79 AND 80

TABLE X: Delete test condition J25 and renumber remaining test conditions.

PAGE 80

TABLE X, NOTE 1, Test condition column: Delete "J1-J42" (six places) and substitute "J1-J41" (six places).

PAGES 98 THROUGH 104

TABLE XVII: Delete and substitute new table XVII as printed on pages 12 through 19 of this amendment.

PAGES 105, 106, AND 107

FIGURES 11a, 11b, and 11c: Delete and substitute new figures 11a and 11b as printed on pages 20 and 21 of this amendment.

PAGE 108

TABLE XVIII, Test condition R12, Max column: Delete "50" and substitute "75".

PAGE 109

TABLE XVIII, Test condition R39, Max column: Delete "50" and substitute "75".

PAGE 111

TABLE XVIII, NOTE 1, Test condition column: Delete "R11" and substitute "R11-R13". Also, delete value and accuracy for test conditions R12 and R13.

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PAGE 112

TABLE XVIII, NOTE 1, Test condition column: Delete "R38", "R39" and "R40-R49" and substitute "R38-R49". Also, delete value and accuracy for test conditions R39 and R40-R49.

Custodians:
Army - ER
Navy - NW
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - NW

(Project 5963-0285)

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6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	MMIO CLK to MMIO REQ	-	-	-	37	37	37	ns	11
2	tpd	MMIO CLK to C/S RG B5	-	-	-	66	66	66	ns	11
3	tpd	EXT OUT EN to C/S RG B5	-	-	-	55	55	55	ns	11
4	tpd	EXT OUT SEL0 to EXT OD B15	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL0 to EXT OD B14	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL0 to EXT OD B13	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL0 to EXT OD B12	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL0 to EXT OD B11	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL0 to EXT OD B10	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL0 to EXT OD B9	-	-	-	86	86	86	ns	11

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tpd	EXT OUT SEL0 to EXT OD B8	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL0 to EXT OD B7	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL0 to EXT OD B6	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL0 to EXT OD B5	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL0 to EXT OD B4	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL0 to EXT OD B3	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL0 to EXT OD B2	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL0 to EXT OD B1	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL0 to EXT OD B0	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL1 to EXT OD B15	-	-	-	86	86	86	ns	11

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tpd	EXT OUT SEL1 to EXT OD B14	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL1 to EXT OD B13	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL1 to EXT OD B12	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL1 to EXT OD B11	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL1 to EXT OD B10	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL1 to EXT OD B9	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL1 to EXT OD B8	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL1 to EXT OD B7	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL1 to EXT OD B6	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL1 to EXT OD B5	-	-	-	86	86	86	ns	11

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tpd	EXT OUT SEL1 to EXT OD B4	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL1 to EXT OD B3	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL1 to EXT OD B2	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL1 to EXT OD B1	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL1 to EXT OD B0	-	-	-	86	86	86	ns	11
5	tpd	EXT OUT EN to EXT OD B0	-	-	-	30	30	30	ns	11
5	tpd	EXT OUT EN to EXT OD B1	-	-	-	30	30	30	ns	11
5	tpd	EXT OUT EN to EXT OD B2	-	-	-	30	30	30	ns	11
5	tpd	EXT OUT EN to EXT OD B3	-	-	-	30	30	30	ns	11
5	tpd	EXT OUT EN to EXT OD B4	-	-	-	30	30	30	ns	11

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
5	tpd	EXT OUT EN to EXT OD B5	-	-	-	30	30	30	ns	11
5	tpd	EXT OUT EN to EXT OD B6	-	-	-	30	30	30	ns	11
5	tpd	EXT OUT EN to EXT OD B7	-	-	-	30	30	30	ns	11
5	tpd	EXT OUT EN to EXT OD B8	-	-	-	30	30	30	ns	11
5	tpd	EXT OUT EN to EXT OD B9	-	-	-	30	30	30	ns	11
5	tpd	EXT OUT EN to EXT OD B10	-	-	-	30	30	30	ns	11
5	tpd	EXT OUT EN to EXT OD B11	-	-	-	30	30	30	ns	11
5	tpd	EXT OUT EN to EXT OD B12	-	-	-	30	30	30	ns	11
5	tpd	EXT OUT EN to EXT OD B13	-	-	-	30	30	30	ns	11
5	tpd	EXT OUT EN to EXT OD B14	-	-	-	30	30	30	ns	11

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
5	tpd	EXT OUT EN to EXT OD B15	-	-	-	30	30	30	ns	11
6	tpd	EXT OD B1 to EXT OD B0	-	-	-	53	53	53	ns	11
7	tpd	LOAD EN to C/S RG B7	-	-	-	33	33	33	ns	11
8	tpd	MMIO CLK to C/S RG B7	-	-	-	75	75	75	ns	11
9	tpd	LOAD EN to C/S RG B6	-	-	-	33	33	33	ns	11
10	tpd	MMIO CLK to C/S RG B6	-	-	-	75	75	75	ns	11
11	tpd	MMIO CLK to RD DAT 0	-	-	-	73	73	73	ns	11
11	tpd	MMIO CLK to RD DAT 1	-	-	-	73	73	73	ns	11
11	tpd	MMIO CLK to RD DAT 2	-	-	-	73	73	73	ns	11
11	tpd	MMIO CLK to RD DAT 3	-	-	-	73	73	73	ns	11

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
11	tpd	MMIO CLK to RD DAT 4	-	-	-	73	73	73	ns	11
11	tpd	MMIO CLK to RD DAT 5	-	-	-	73	73	73	ns	11
11	tpd	MMIO CLK to RD DAT 6	-	-	-	73	73	73	ns	11
11	tpd	MMIO CLK to RD DAT 7	-	-	-	73	73	73	ns	11
11	tpd	MMIO CLK to RD DAT 8	-	-	-	73	73	73	ns	11
11	tpd	MMIO CLK to RD DAT 9	-	-	-	73	73	73	ns	11
11	tpd	MMIO CLK to RD DAT 10	-	-	-	73	73	73	ns	11
11	tpd	MMIO CLK to RD DAT 11	-	-	-	73	73	73	ns	11
11	tpd	MMIO CLK to RD DAT 12	-	-	-	73	73	73	ns	11
11	tpd	MMIO CLK to RD DAT 13	-	-	-	73	73	73	ns	11

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
11	tpd	MMIO CLK to RD DAT 14	-	-	-	73	73	73	ns	11
11	tpd	MMIO CLK to RD DAT 15	-	-	-	73	73	73	ns	11
12	tpd	MMIO CLK to RD DAT 0	-	-	-	73	73	73	ns	11
12	tpd	MMIO CLK to RD DAT 1	-	-	-	73	73	73	ns	11
12	tpd	MMIO CLK to RD DAT 2	-	-	-	73	73	73	ns	11
12	tpd	MMIO CLK to RD DAT 3	-	-	-	73	73	73	ns	11

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TABLE XVII. Dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	MMIO CLK to MMIO REQ	-	-	-	37	37	37	ns	11
2	tpd	MMIO CLK to C/S RG B5	-	-	-	66	66	66	ns	11
3	tpd	EXT OUT EN to C/S RG B5	-	-	-	55	55	55	ns	11
4	tpd	EXT OUT SEL0 to EXT OD B15	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL0 to EXT OD B14	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL0 to EXT OD B13	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL0 to EXT OD B12	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL0 to EXT OD B11	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL0 to EXT OD B10	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL0 to EXT OD B9	-	-	-	86	86	86	ns	11

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tpd	EXT OUT SEL0 to EXT OD B8	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL0 to EXT OD B7	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL0 to EXT OD B6	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL0 to EXT OD B5	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL0 to EXT OD B4	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL0 to EXT OD B3	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL0 to EXT OD B2	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL0 to EXT OD B1	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL0 to EXT OD B0	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL1 to EXT OD B15	-	-	-	86	86	86	ns	11

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tpd	EXT OUT SEL1 to EXT OD B14	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL1 to EXT OD B13	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL1 to EXT OD B12	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL1 to EXT OD B11	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL1 to EXT OD B10	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL1 to EXT OD B9	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL1 to EXT OD B8	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL1 to EXT OD B7	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL1 to EXT OD B6	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL1 to EXT OD B5	-	-	-	86	86	86	ns	11

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tpd	EXT OUT SEL1 to EXT OD B4	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL1 to EXT OD B3	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL1 to EXT OD B2	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL1 to EXT OD B1	-	-	-	86	86	86	ns	11
4	tpd	EXT OUT SEL1 to EXT OD B0	-	-	-	86	86	86	ns	11
5	tpd	EXT OUT EN to EXT OD B0	-	-	-	30	30	30	ns	11
5	tpd	EXT OUT EN to EXT OD B1	-	-	-	30	30	30	ns	11
5	tpd	EXT OUT EN to EXT OD B2	-	-	-	30	30	30	ns	11
5	tpd	EXT OUT EN to EXT OD B3	-	-	-	30	30	30	ns	11
5	tpd	EXT OUT EN to EXT OD B4	-	-	-	30	30	30	ns	11

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
5	tpd	EXT OUT EN to EXT OD B5	-	-	-	30	30	30	ns	11
5	tpd	EXT OUT EN to EXT OD B6	-	-	-	30	30	30	ns	11
5	tpd	EXT OUT EN to EXT OD B7	-	-	-	30	30	30	ns	11
5	tpd	EXT OUT EN to EXT OD B8	-	-	-	30	30	30	ns	11
5	tpd	EXT OUT EN to EXT OD B9	-	-	-	30	30	30	ns	11
5	tpd	EXT OUT EN to EXT OD B10	-	-	-	30	30	30	ns	11
5	tpd	EXT OUT EN to EXT OD B11	-	-	-	30	30	30	ns	11
5	tpd	EXT OUT EN to EXT OD B12	-	-	-	30	30	30	ns	11
5	tpd	EXT OUT EN to EXT OD B13	-	-	-	30	30	30	ns	11
5	tpd	EXT OUT EN to EXT OD B14	-	-	-	30	30	30	ns	11

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
5	tpd	EXT OUT EN to EXT OD B15	-	-	-	30	30	30	ns	11
6	tpd	EXT OD B1 to EXT OD B0	-	-	-	53	53	53	ns	11
7	tpd	LOAD EN to C/S RG B7	-	-	-	33	33	33	ns	11
8	tpd	MMIO CLK to C/S RG B7	-	-	-	75	75	75	ns	11
9	tpd	LOAD EN to C/S RG B6	-	-	-	33	33	33	ns	11
10	tpd	MMIO CLK to C/S RG B6	-	-	-	75	75	75	ns	11
11	tpd	MMIO CLK to RD DAT 0	-	-	-	73	73	73	ns	11
11	tpd	MMIO CLK to RD DAT 1	-	-	-	73	73	73	ns	11
11	tpd	MMIO CLK to RD DAT 2	-	-	-	73	73	73	ns	11
11	tpd	MMIO CLK to RD DAT 3	-	-	-	73	73	73	ns	11

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
11	tpd	MMIO CLK to RD DAT 4	-	-	-	73	73	73	ns	11
11	tpd	MMIO CLK to RD DAT 5	-	-	-	73	73	73	ns	11
11	tpd	MMIO CLK to RD DAT 6	-	-	-	73	73	73	ns	11
11	tpd	MMIO CLK to RD DAT 7	-	-	-	73	73	73	ns	11
11	tpd	MMIO CLK to RD DAT 8	-	-	-	73	73	73	ns	11
11	tpd	MMIO CLK to RD DAT 9	-	-	-	73	73	73	ns	11
11	tpd	MMIO CLK to RD DAT 10	-	-	-	73	73	73	ns	11
11	tpd	MMIO CLK to RD DAT 11	-	-	-	73	73	73	ns	11
11	tpd	MMIO CLK to RD DAT 12	-	-	-	73	73	73	ns	11
11	tpd	MMIO CLK to RD DAT 13	-	-	-	73	73	73	ns	11

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
11	tpd	MMIO CLK to RD DAT 14	-	-	-	73	73	73	ns	11
11	tpd	MMIO CLK to RD DAT 15	-	-	-	73	73	73	ns	11
12	tpd	MMIO CLK to RD DAT 0	-	-	-	73	73	73	ns	11
12	tpd	MMIO CLK to RD DAT 1	-	-	-	73	73	73	ns	11
12	tpd	MMIO CLK to RD DAT 2	-	-	-	73	73	73	ns	11
12	tpd	MMIO CLK to RD DAT 3	-	-	-	73	73	73	ns	11

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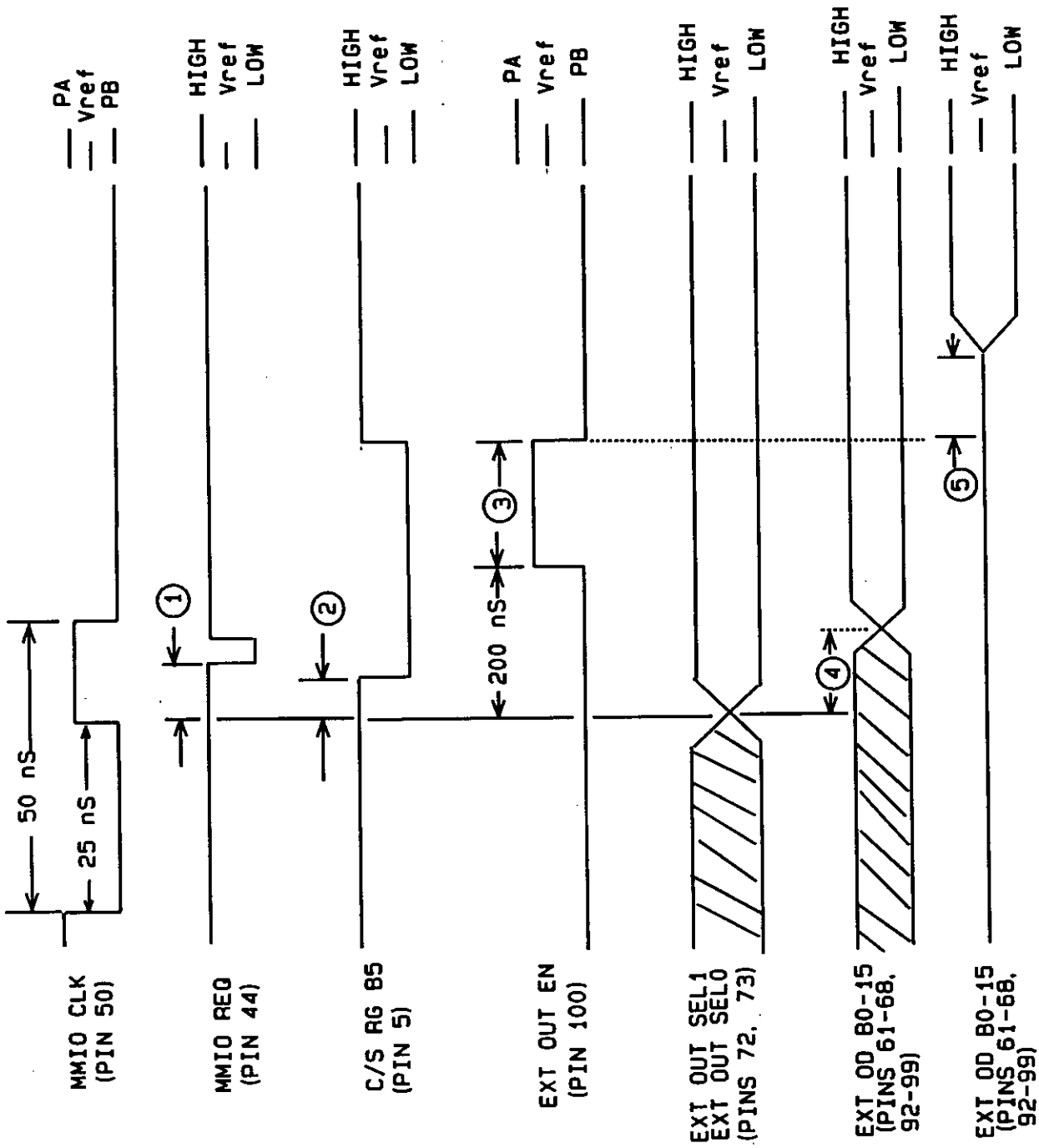


FIGURE 11a. Dynamic waveforms for table XVII.

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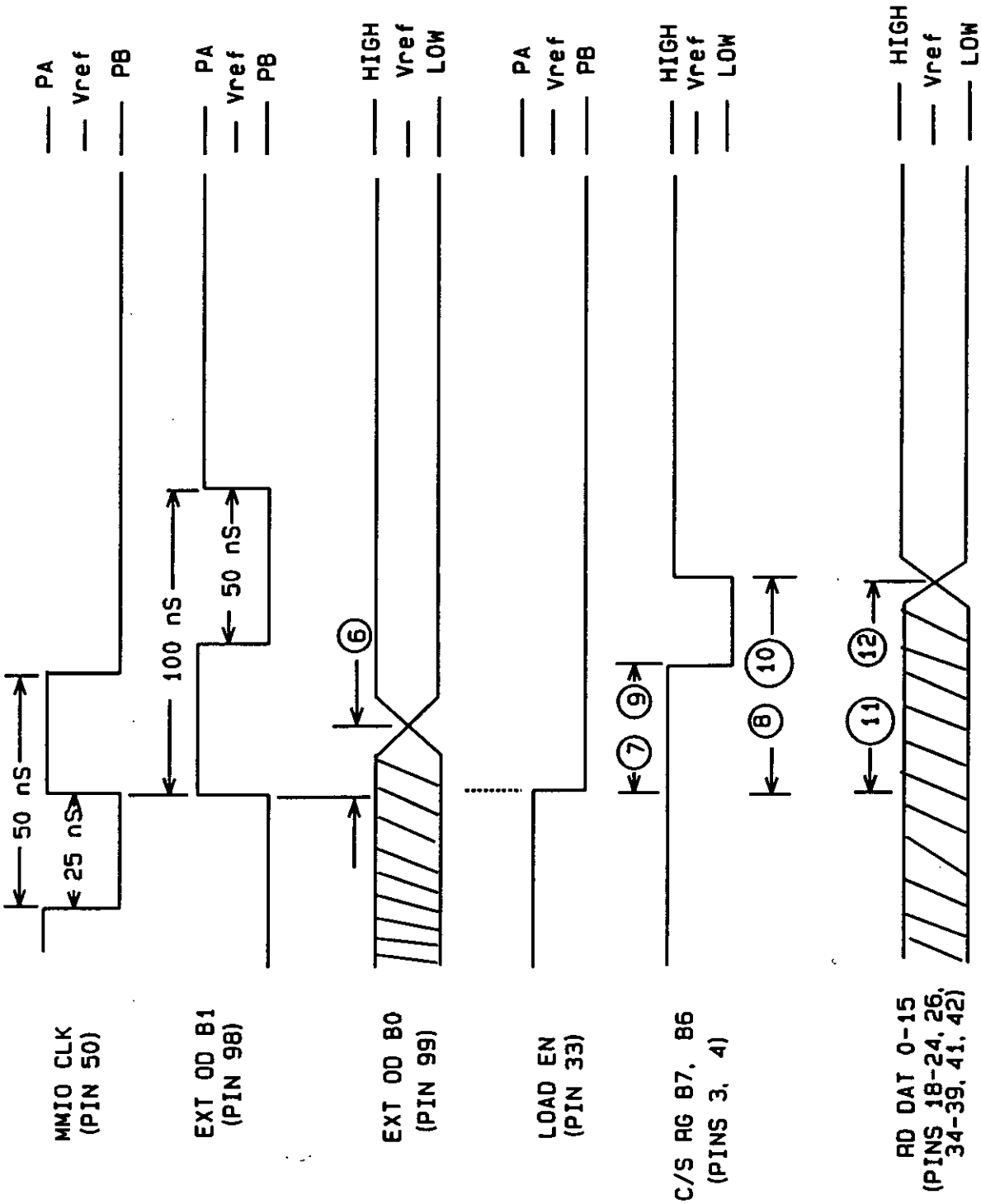


FIGURE 11b. Dynamic waveforms for table XVII.

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